Electrical and Computer Engineering
University of Thessaly

ECE214 - Introduction to Electronics

Characterization of CMOS Combinational Gates and Full Adder Circuits via SPICE Simulations

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Assignment 4

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Contents

Exercise 6.1 2

Exercise 6.2 4

Exercise 7.1 8

Exercise 7.2 8

Exercise 6.1

CMOS NAND Propagation Delay Analysis

The transient performance of the CMOS NAND gate was evaluated by analyzing the output voltage waveforms during distinct input vector transitions. Initially, when the input state switches from $(A, B) = (1, 1)$ to $(0, 0)$, the output undergoes a low-to-high transition. The corresponding propagation delay for this state change was determined to be $t_{pLH} = 295.11$ ps, as illustrated in Figure 1.

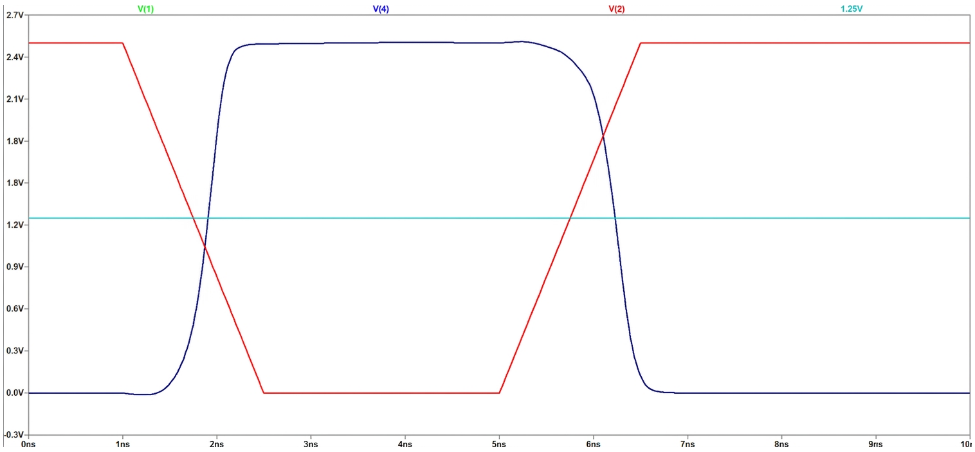

Figure 1: SPICE transient response and propagation delay t_{pLH} for the input vector transition $(A, B) = (1, 1) \rightarrow (0, 0)$.

In a similar manner, the remaining low-to-high output transitions were evaluated. When the input vector switches from $(A, B) = (1, 1)$ to $(1, 0)$, the measured propagation delay is $t_{pLH} = 595.7$ ps. Furthermore, the input transition from $(1, 1)$ to $(0, 1)$ yields a propagation delay of $t_{pLH} = 534.38$ ps.

Regarding the high-to-low output response, when the input vector changes from $(0, 0)$ to $(1, 1)$, the output transitions from a high logic level (logic 1) to a low logic level (logic 0). For this specific operation, the propagation delay was determined to be $t_{pHL} = 661.84$ ps (figure 2). It is crucial to denote that at this state, the internal capacitance of the pull-down network is initially uncharged as we can notice in figure 2 where the characteristic of V(3) is mentained below a certain level . To provide a comprehensive analysis, the propagation delay under a pre-charged internal capacitance condition will be subsequently calculated and discussed.

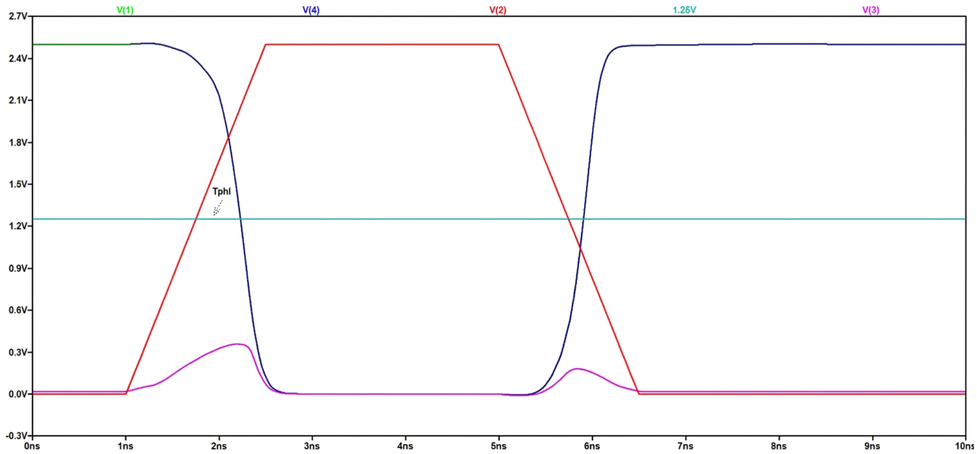

Figure 2: SPICE transient response and propagation delay t_{pHL} for the input vector transition $(A, B) = (0, 0) \rightarrow (1, 1)$.

In a similar manner, the remaining high-to-low output transitions were analyzed. Specifically, when the input vector switches from $(1, 0)$ to $(1, 1)$, the propagation delay is measured at $t_{pHL} = 554.4$ ps. Conversely, when the input transitions from $(0, 1)$ to $(1, 1)$, the resulting high-to-low propagation delay is equal to $t_{pHL} = 538.51$ ps.

Based on these simulation results, it is observed that the propagation delay during the $(0, 0) \rightarrow (1, 1)$ transition is significantly larger compared to any other high-to-low output transition. This behavior occurs when the internal capacitance of the pull-down network is initially uncharged.

The propagation delay becomes even larger under a pre-charged internal capacitance condition. To evaluate this worst-case scenario, the internal node capacitor was pre-charged by sequentially switching the input state from $(1, 0)$ to $(0, 0)$, and subsequently transitioning to the final input vector of $(1, 1)$ as observed in figure 3.

More specifically ,in figure 3 someone can distinguish the corresponding simulation diagram illustrating the input and output voltage waveforms, along with the voltage across the internal node capacitor which is pre-charged prior to the final input transition.

3.

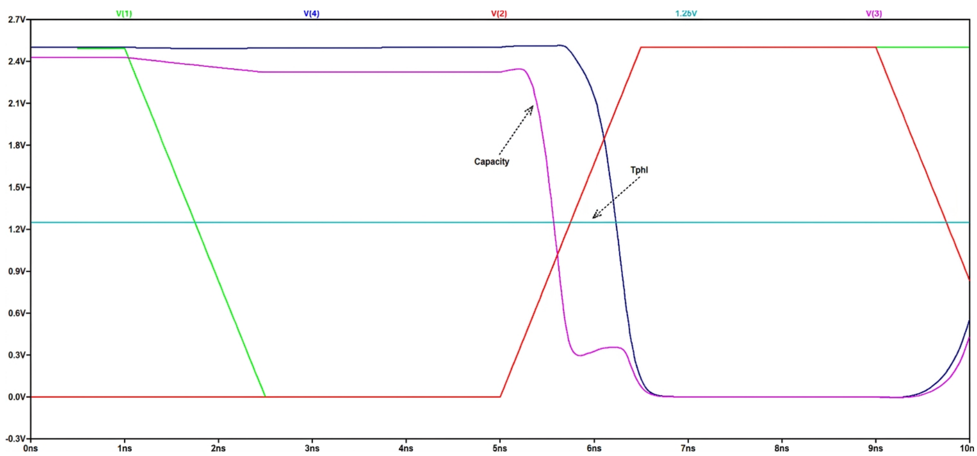

Figure 3: SPICE transient response and propagation delay t_{pHL} for the input vector transition $(A, B) = (0, 0) \rightarrow (1, 1)$ with a pre charged capacitor.

By measuring the propagation delay we estimate the time $t_{phl} = 676.55$ ps, which is significantly larger compared to the scenario where the same capacitor was initially uncharged.

In summary, the measured propagation delays for all examined input vector transitions are compiled in Table 1.

Table 1: Summary of CMOS NAND Gate Propagation Delays.

Input Transition	Output Transition	Delay Parameter	Value (ps)
$(1,1) \rightarrow (0,0)$	Low $\rightarrow$ High	t_{pLH}	295.11
$(1,1) \rightarrow (1,0)$	Low $\rightarrow$ High	t_{pLH}	595.70
$(1,1) \rightarrow (0,1)$	Low $\rightarrow$ High	t_{pLH}	534.38
$(0,0) \rightarrow (1,1)$ (uncharged)	High $\rightarrow$ Low	t_{pHL}	661.84
$(1,0) \rightarrow (1,1)$	High $\rightarrow$ Low	t_{pHL}	554.40
$(0,1) \rightarrow (1,1)$	High $\rightarrow$ Low	t_{pHL}	538.51
$(0,0) \rightarrow (1,1)$ (pre-charged)	High $\rightarrow$ Low	t_{pHL}	676.55

Based on the comprehensive simulation results, the worst-case input transition causing the maximum low-to-high propagation delay occurs during the $(1,1) \rightarrow (1,0)$ transition, yielding $t_{pLH} = 595.7$ ps. Conversely, the worst-case scenario for the high-to-low output response is observed during the $(0,0) \rightarrow (1,1)$ transition when the internal node capacitance of the pull-down network is pre-charged, resulting in the absolute maximum propagation delay of $t_{pHL} = 676.55$ ps.

Exercise 6.2

CMOS Complex Gate Analysis

In this exercise, a complex CMOS logic simulating the behaviour of a full adder gate needs to be evaluated and analyzed . The schematic circuit topology, structured with the appropriate pull-up and pull-down networks, is illustrated in Figure 4.

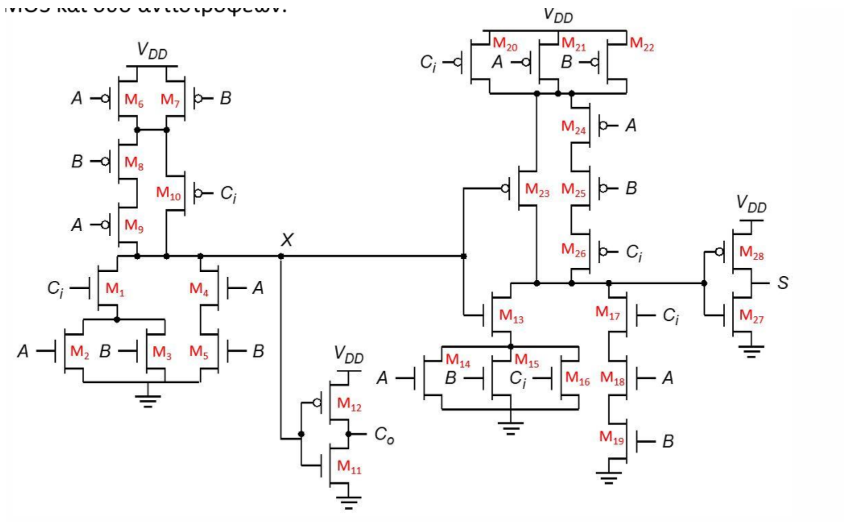

Figure 4: Schematic diagram of 1-bit full-adder.

First complex gate

By inspecting the topology and wiring of the Pull-Up Network (PUN) consisting of pMOS transistors we have the following boolean expressions:

$$\begin{aligned}(A' + B') \cdot (B' \cdot A' + C_i') &= A'A'B' + A'C_i' + B'B'A' + B'C_i' \\ &= A'B' + A'C_i' + B'A' + B'C_i' \\ &= A'B' + A'C_i' + B'C_i' = X\end{aligned}$$

Similarly, the Pull-Down Network (PDN), built exclusively with nMOS transistors, is described by the following Boolean expression:

$$\begin{aligned}[C_i \cdot (A + B) + A \cdot B]' &= (C_i' + A' \cdot B')(A' + B') \\ &= (A' + B')(C_i' + A' \cdot B') \\ &= A'C_i' + B'C_i' + A'A'B' + A'B'B' \\ &= A'C_i' + B'C_i' + A'B' + A'B' \\ &= A'C_i' + B'C_i' + A'B' = X\end{aligned}$$

So both PUN and PDN verify the fact the the first CMOS gate is described by the following boolean expression :$X = A'B' + A'C_i' + B'C_i'$

Second complex gate

By inspecting the topology of the PUN we have the following mathematical description:

$$
\begin{aligned}
(C_i' + A' + B') \cdot (X' + A'B'C_i') &= C_i'X' + C_i'A'B'C_i' + A'X' + A'A'B'C_i' + B'X' + B'A'B'C_i' \\
' &= C_i'X' + A'B'C_i' + A'X' + A'B'C_i' + B'X' + A'B'C_i' \\
&= C_i'X' + A'B'C_i' + A'X' + B'X' \\
&= X' \cdot (C_i' + A' + B') + A'B'C_i' \\
&= [(A+B) \cdot (A+C_i) \cdot (B+C_i)] \cdot (C_i' + A' + B') + A'B'C_i' \\
&= [(AA + AC_i + AB + BC_i) \cdot (B+C_i)] \cdot (C_i' + A' + B') + A'B'C_i' \\
&= [AB + AC_i + ABC_i + AC_i + AB + ABC_i + BC_i] \cdot (C_i' + A' + B') + A \\
&= [AB + AC_i + BC_i + ABC_i] \cdot (C_i' + A' + B') + A'B'C_i' \\
&= ABC_i' + AB'C_i + A'BC_i' + A'B'C_i' = F
\end{aligned}
$$

Similarly for PDN we have the following function:

$$
\begin{aligned}
&[(A + B + C_i) \cdot X + ABC_i]' \\
&\qquad = A'B'C_i' + (X' \cdot (A' + B' + C_i') = \cdots \\
&\qquad = ABC_i' + AB'C_i + A'BC_i' + A'B'C_i' = F
\end{aligned}
$$

The boolean algebra verifies this time that the second CMOS gate is described by the following boolean function:$F = ABC_i' + AB'C_i + A'BC_i' + A'B'C_i'$

The truth table of the Full Adder is:

A	**B**	**Ci**	**Co**	**S**	**Co'**	**S'**
0	0	0	0	0	1	1
0	0	1	0	1	1	0
0	1	0	0	1	1	0
0	1	1	1	0	0	1
1	0	0	0	1	1	0
1	0	1	1	0	0	1
1	1	0	1	0	0	1
1	1	1	1	1	0	0

By using the Karnaugh map, we have:

$$C_o' = A'B' + A'C_i' + B'C_i' = X$$

$$S' = A'B'C_i' + A'BC_i + AB'C_i + ABC_i' = F$$

Since the logical functions implemented by the two complex gates actually identical to the complements of the two outputs of a Full Adder, the operation of the circuit as a Full Adder is confirmed.

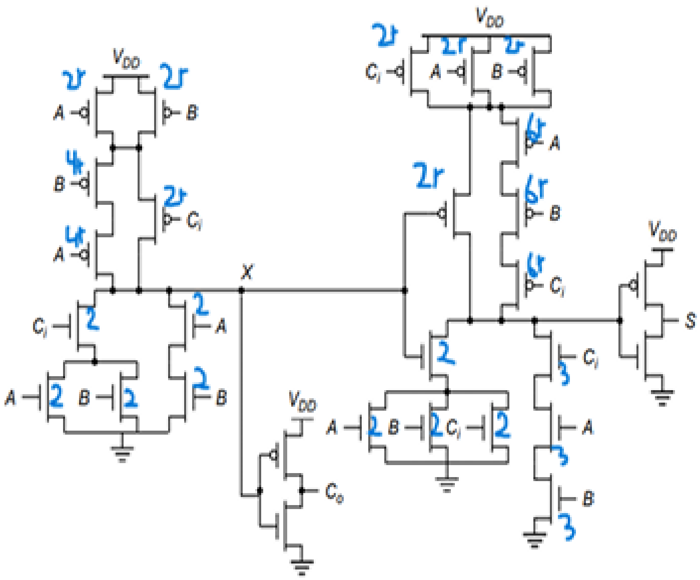

Figure 5: Schematic diagram of 1-bit full-adder with the widths of each transistor.

For the widths of each transistor of the entire full adder implementation we have the following designing (figure 5):

Where $r = 4.81$ (k'_n/k'_p).

The gate capacitance of the load transistors connected to the output of the complex gate is calculated as follows:

$$\begin{aligned}
C_{L(\text{ext})} &= C_{g1} + C_{g2} + C_{g3} + C_{g4} \\
&= \frac{\varepsilon_{ox}}{t_{ox}} W_1 L + \frac{\varepsilon_{ox}}{t_{ox}} W_2 L + \frac{\varepsilon_{ox}}{t_{ox}} W_3 L + \frac{\varepsilon_{ox}}{t_{ox}} W_4 L \\
&= \frac{\varepsilon_{ox}}{t_{ox}} L (W_1 + W_2 + W_3 + W_4) \\
&= \frac{4 \cdot 8.85 \cdot 10^{-14} \cdot 100 \cdot 0.25 \cdot 10^{-6}}{5.7 \cdot 10^{-9}} \cdot (2r + r + 1 + 2) \cdot 10^{-6} \\
&= 27.06 \cdot 10^{-15}\ \text{F} = 0.02706 \cdot 10^{-12}\ \text{F} = 0.027\ \text{pF}
\end{aligned}$$

By using SPICE, we represent the voltage diagrams for each case of low-to-high output transition and vice versa. More specifically:

The input changes from $(0, 0, 0)$ to $(1, 1, 1)$. The output of the first complex gate of the full adder (i.e., the complement of the carry out) transitions from logic 1 to logic 0. The propagation delay was measured at $t_{pHL} = 568.42$ ps as we observe in figure 7.

Similar are the remaining transition diagrams for the high-to-low output level are . According to spice measurements an input transition from $(0, 0, 0)$ to $(1, 1, 0)$, could cause a propagation delay equal with $t_{pHL} = 1606$ ps.

If the input changes from $(0, 0, 0)$ to $(1, 0, 1)$, then the propagation delay is equal to $t_{pHL} = 911.14$ ps.

While if the input changes from $(0, 0, 0)$ to $(0, 1, 1)$, then the propagation delay becomes equal to $t_{pHL} = 1474$ ps.

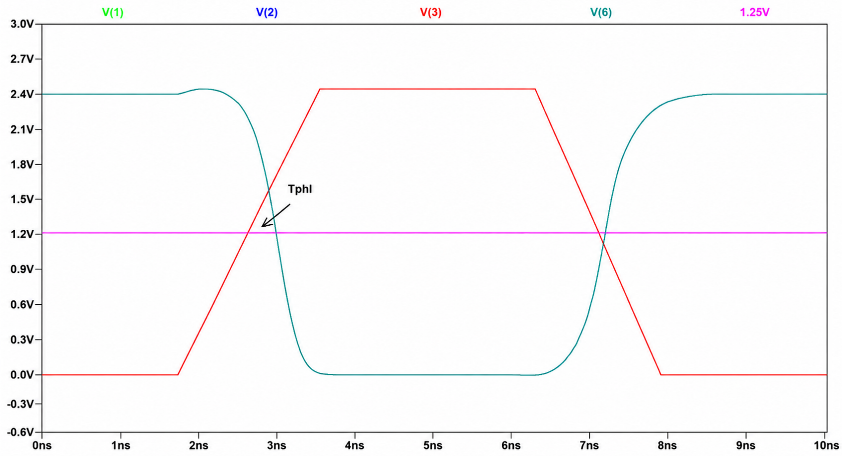

Figure 6: SPICE transient response and propagation delay $t_{pHL}$for the input vector transition (A, B,Ci) = (0, 0,0) → (1, 1, 1)

In the following diagram (figure **??**) , we observe that if the input changes from $(1, 1, 1)$ to $(0, 0, 0)$, the output transitions from a low voltage level to a high level. The propagation delay was measured at $t_{pLH} = 335.52$ ps.

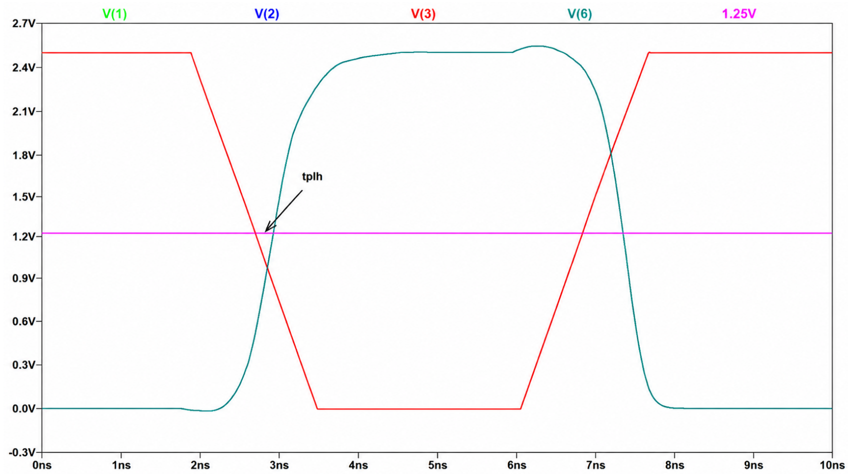

Figure 7: SPICE transient response and propagation delay $t_{pLH}$for the input vector transition (A, B,Ci) = (1, 1,1) → (0, 0, 0)

The remaining transition diagrams for the low-to-high output level are showing a g a similar behaviour .

If the input changes from $(1, 1, 1)$ to $(0, 0, 1)$, then the propagation delay is $t_{pLH} = 551.07$ ps.

If the input changes from $(1, 1, 1)$ to $(0, 1, 0)$, then the propagation delay is $t_{pLH} = 405.91$ ps.

While if the input changes from $(1, 1, 1)$ to $(1, 0, 0)$, then the propagation delay becomes equal to $t_{pLH} = 632.78$ ps.

The table 2 summarizes all the measurements that have been performed:

Table 2: Summary of CMOS Complex Gate Propagation Delays.

Input Transition	Output Transition	Delay Parameter	Value (ps)
$(0,0,0) \to (1,1,1)$	High $\to$ Low	t_{pHL}	568.42
$(0,0,0) \to (1,1,0)$	High $\to$ Low	t_{pHL}	1606.00
$(0,0,0) \to (1,0,1)$	High $\to$ Low	t_{pHL}	911.14
$(0,0,0) \to (0,1,1)$	High $\to$ Low	t_{pHL}	1474.00
$(1,1,1) \to (0,0,0)$	Low $\to$ High	t_{pLH}	335.52
$(1,1,1) \to (0,0,1)$	Low $\to$ High	t_{pLH}	551.07
$(1,1,1) \to (0,1,0)$	Low $\to$ High	t_{pLH}	405.91
$(1,1,1) \to (1,0,0)$	Low $\to$ High	t_{pLH}	632.70

Exercise 7.1

Power and Energy Analysis of a Symmetric Inverter

The average power dissipation over a time period T is given by the following analytical formula:

$$P_T = \frac{1}{T}\int_0^T P(t)\,dt = \frac{1}{T}\int_0^T V_{DD}\cdot I(t)_{DD}\,dt = \frac{V_{DD}}{T}\int_0^T I(t)_{DD}\,dt$$

By considering the supply voltage to be stable and equal to $V_{DD} = 2.5$ V and the time period $T = 30$ ns, we simulate the behavior of a symmetric inverter for an input transition of $1 \to 0 \to 1$ with rise and fall times set at $t_r = t_f = 7.5$ ns, as observed in Figure 8.

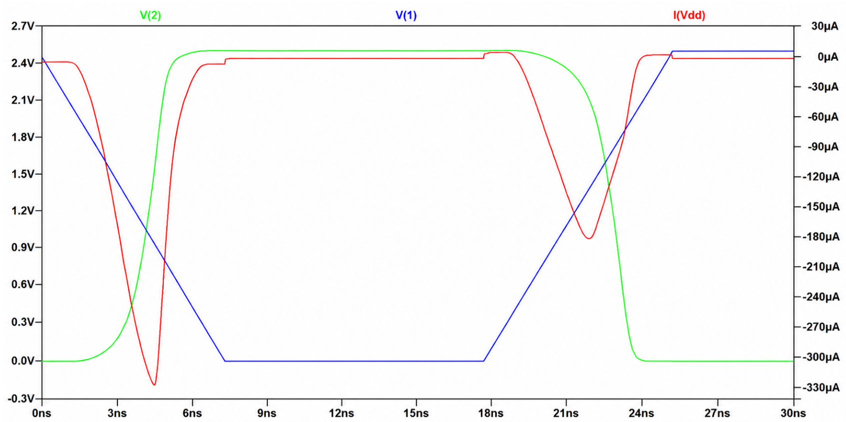

Figure 8:

By using SPICE transient simulation, the average power dissipation is determined to be:

$$P_T = 9.593 \cdot 10^{-5}\ \mathrm{W}$$

The total energy delivered by the power supply is given by the following relation:

$$E = \int_0^T P(t)\,dt = \int_0^T V_{DD}\cdot I(t)_{DD}\,dt = V_{DD}\int_0^T I(t)_{DD}\,dt$$

And by using SPICE, we conclude that the total energy is:

$$E = 2.877 \cdot 10^{-12}\ \mathrm{J}$$

Exercise 7.2

Minimum Input Transition Time Analysis

After conducting multiple SPICE simulations, it was concluded that the minimum time interval between two consecutive input transitions for which the gate operates sufficiently correct is estimated to be 2 ns.

More specifically, as shown below, during the input transition from $(0, 1, 0)$ to $(0, 1, 1)$, the output of the complex gate transitions from a high to a low voltage level. However, it fails to completely reach the 0 V rail because the subsequent input vector forces it to rise again prematurely.Although the output transition from a high voltage level to a low one is not perfect, since the voltage levels do not reach absolute zero during the period between two consecutive input changes ,if we consider a time period of 2 ns its behavior is considered barely functional,thus the minimum time is set for this time period.

Furthermore, an inverter has been added to the circuit configuration, which essentially generates the true carry-out value (C_o) for every input combination of a Full Adder. It is observed that its output response is sufficiently stable and well-defined for $T = 2$ ns. Consequently, the required minimum transition time is confirmed to be 2 ns.The output dynamic behaviour for both the first CMOS gate as well as its invrtered output as presented in figure 9

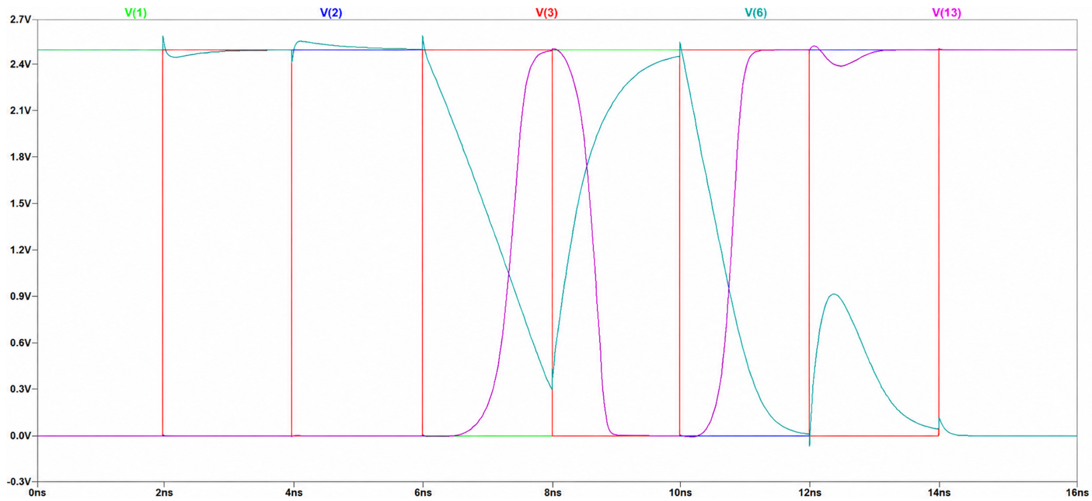

Figure 9: Output Characteristics for Each Input Transitions of the First CMOS Gate

By using SPICE simulation, we find that the average power dissipation of the circuit for this minimum transition time of 2 ns is equal to:

$$P_{\text{avg}} = 0.000445 \text{ W} = 0.44 \text{ mW}$$

For the theoretical calculation of the average power consumption, it is required to determine the output switching activity factor of the gate. Additionally, it must be taken into consideration that the total capacitance utilized in the subsequent power calculation formula must include the internal diffusion (junction) capacitances of the transistor regions connected to node X. Specifically, we are interested in the following transistors in figure 10:

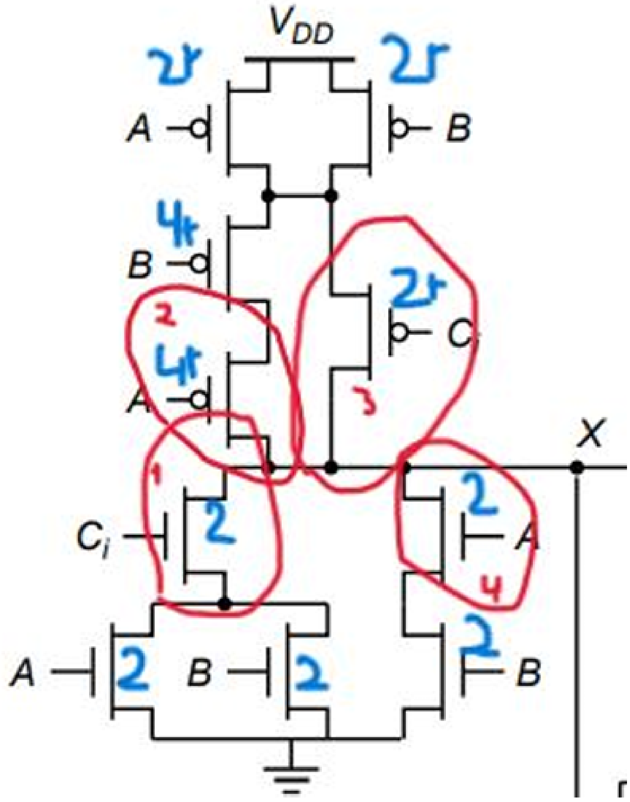

Figure 10: Transistors of the first gate whose junction and perimeter capacitances must be calculated.

In general, the junction area capacitance (C_{jar}) and the sidewall perimeter capacitance (C_{jsw}) are given by the following formulas:

$$C_{jar} = \overline{C_{jar}} \cdot A_D$$
$$C_{jsw} = \overline{C_{jsw}} \cdot P_D$$

Where $A_D = W \cdot L_D$ represents the drain junction area and $P_D = W + 2L_D$ represents the drain junction perimeter.

Additionally, the technology parameters $\overline{C_{jar}}$ and $\overline{C_{jsw}}$ are specified within the SPICE transistor models and vary depending on the type of the device (PMOS or NMOS).

Specifically, for Transistor 1 (nMOS transistor), the calculated capacitance components are:

$$C_{jar} = \overline{C_{jar}} \cdot A_D = 0.0181 \text{ pF}$$
$$C_{jsw} = \overline{C_{jsw}} \cdot P_D = 0.0064 \text{ pF}$$

Similarly, for Transistor 2 (pMOS transistor) of the aforementioned schematic diagram, we obtain:

$$C_{jar} = \overline{C_{jar}} \cdot A_D = 0.18 \text{ pF}$$
$$C_{jsw} = \overline{C_{jsw}} \cdot P_D = 0.01 \text{ pF}$$

Similarly, for Transistor 3 (pMOS transistor) of the aforementioned schematic diagram, we obtain:

$$C_{jar} = \overline{C_{jar}} \cdot A_D = 0.09 \text{ pF}$$
$$C_{jsw} = \overline{C_{jsw}} \cdot P_D = 0.0071 \text{ pF}$$

For Transistor 4 (nMOS transistor) of the same schematic, the capacitance components are calculated as:

$$C_{jar} = \overline{C_{jar}} \cdot A_D = 0.018 \text{ pF}$$
$$C_{jsw} = \overline{C_{jsw}} \cdot P_D = 0.0064 \text{ pF}$$

By summing the above junction and sidewall capacitances, we derive the total internal parasitic capacitance contribution at node X:

$$C_{\text{in}} = 0.333 \text{ pF}$$

Consequently, the total capacitance that interests us, which includes both the internal parasitic contribution (C_{in}) and the external load capacitance (C_{out}), is given by:

$$C_L = C_{\text{in}} + C_{\text{out}} = 0.36 \text{ pF}$$

By using the analytical formula for dynamic power consumption:

$$P_{av} = \alpha_{0\to 1} \cdot f \cdot C_L \cdot V_{DD}^2$$

Where the individual parameters are defined as follows:

$$\begin{aligned} \alpha_{0\to 1} &= \frac{1}{8} \\ f &= \frac{1}{T} = 0.0625 \cdot 10^9 \text{ Hz} = 62.5 \text{ MHz} \\ C_L &= 0.36 \text{ pF} \\ V_{DD} &= 2.5 \text{ V} \end{aligned}$$

Substituting these values into the equation yields:

$$P_{av} = 0.000017 \text{ W} = 0.17 \text{ mW}$$

Consequently, we observe that the theoretically calculated average power consumption is significantly lower than the experimental average power derived from the SPICE transient simulation (0.44 mW).

This discrepancy is attributed to two additional parameters that are not taken into account by the aforementioned analytical formula, whereas SPICE considers them: the short-circuit current and the leakage current.

Regarding the short-circuit current, during a switching transition in a static CMOS gate, there is a probability that both the PMOS and NMOS transistors are simultaneously turned on for a short time interval. This interval depends on the rise and fall times of the input and output signals. During this period, a direct current path is formed between V_{DD} and ground. The power delivered by this current is dissipated in the transistors without contributing to the charging of the output load capacitance; this power loss occurs only when the output changes state. The short-circuit current depends on the value of V_{DD}. If $V_{DD} < V_{TN} + |V_{TP}|$, then the transistors are never simultaneously turned on, and thus this phenomenon is eliminated.

However, even when $V_{GS} < V_T$, a leakage current still flows through the transistor, a quantity that continuously increases with technology scaling. Consequently, the circuit exhibits power consumption even in the absence of output switching transitions at the gate.

Leakage currents depend significantly on the supply voltage of the circuit. Furthermore, they are dependent on temperature and inversely dependent on the transistor's threshold voltage (V_T); in fact, this dependence is exponential. However, this reduction in threshold voltage inherently leads to larger currents, which means that the output capacitances charge faster, thereby increasing the overall speed of the circuit. As a result, smaller technology nodes can be more efficient in terms of speed, but they suffer from increased static power consumption.